

Lectures 34 & 35 — Graphs, Degree, and the Handshaking Lemma

CS 173: Discrete Structures · Graph Theory

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1 Defining a graph

A graph is a combinatorial object built from two ingredients: a collection of *points* (called vertices or nodes) and a collection of *connections* between pairs of those points (called edges). Both ingredients are sets, the vertex set is required to be finite, and the only structural constraint is that every edge is a 2-element subset of the vertex set.

Definition 1.1 (Graph). A **graph** G is determined by

- a *finite* set of *nodes* (or *vertices*) $V(G)$, and
- a set of *edges* $E(G)$,

such that

$$E(G) \subseteq \{W \mid W \subseteq V(G) \wedge |W| = 2\}.$$

The symbol G in $V(G)$ and $E(G)$ is the *name of the graph*; the functions $V(\cdot)$ and $E(\cdot)$ extract the vertex set and edge set of whichever graph you name. Read $V(G)$ as “the vertex set of G ” and $E(G)$ as “the edge set of G .”

Remark. The condition $E(G) \subseteq \{W \mid W \subseteq V(G) \wedge |W| = 2\}$ is doing a lot of work. Unpacking it:

- Each edge is a *set* W , not an ordered pair, so $\{0, 1\} = \{1, 0\}$ denotes a single edge. This is the formal reason these graphs are **undirected**.
- $W \subseteq V(G)$ forces both endpoints to be vertices of G .
- $|W| = 2$ forces the two endpoints to be *distinct*, ruling out loops $\{v, v\} = \{v\}$ (which has cardinality 1, not 2).
- $E(G)$ is itself a set, so no edge is repeated — this rules out parallel edges (multigraphs).

Finiteness of $V(G)$ then forces $E(G)$ to be finite as well, since $E(G) \subseteq \mathcal{P}(V(G))$ and the power set of a finite set is finite.

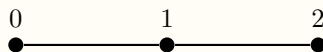
Caution. A graph is determined by the *pair* $(V(G), E(G))$, not by $E(G)$ alone. Two graphs can have the same edge set but different vertex sets if one of them has *isolated* vertices — vertices in $V(G)$ that appear in no edge.

2 First example

Example. Let G be the graph with

$$V(G) = \{0, 1, 2\}, \quad E(G) = \{\{0, 1\}, \{1, 2\}\}.$$

Each element of $E(G)$ is a 2-element subset of $V(G)$, so the definition is satisfied.



The vertex 1 is shared by both edges; vertices 0 and 2 each belong to exactly one edge, and there is no edge directly between 0 and 2.

3 Neighbourhood and incident edges

Once a graph G and a vertex $v \in V(G)$ are fixed, two natural “local” sets describe how v sits inside G : the other vertices it is joined to, and the edges that touch it. The first lives in $V(G)$, the second in $E(G)$.

Definition 3.1 (Neighbourhood). Given a graph G and a node $v \in V(G)$, the **neighbourhood** of v in G is

$$N_G(v) := \{w \in V(G) \mid \{v, w\} \in E(G)\}.$$

That is, $N_G(v)$ is the set of vertices joined to v by an edge.

Definition 3.2 (Incident edges). Given a graph G and a node $v \in V(G)$, the set of **edges incident to** v is

$$I_G(v) := \{e \in E(G) \mid v \in e\}.$$

That is, $I_G(v)$ is the set of edges that have v as one of their two endpoints.

The two sets answer different questions, and the type-checking matters:

Set	Element type	Lives in
$N_G(v)$	a <i>vertex</i> w adjacent to v	$\mathcal{P}(V(G))$
$I_G(v)$	an <i>edge</i> e containing v	$\mathcal{P}(E(G))$

The two are tightly linked: each $e \in I_G(v)$ has the form $e = \{v, w\}$ for exactly one $w \in N_G(v)$, and conversely each $w \in N_G(v)$ corresponds to a unique edge $\{v, w\} \in I_G(v)$. So $|N_G(v)| = |I_G(v)|$ for every vertex.

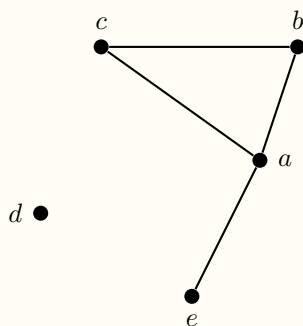
Caution. $v \notin N_G(v)$. The neighbourhood of v never contains v itself, because $\{v, v\} = \{v\}$ has cardinality 1, so it is not a valid edge. Some texts use $N_G[v] = N_G(v) \cup \{v\}$ for the *closed* neighbourhood; the version on the board is the *open* neighbourhood and excludes v .

4 Worked example

The board showed a five-vertex graph G to compute N_G and I_G on a less trivial picture. Let

$$V(G) = \{a, b, c, d, e\}, \quad E(G) = \{\{a, b\}, \{a, c\}, \{a, e\}, \{b, c\}\}.$$

The vertex d is in $V(G)$ but is not an endpoint of any edge — it is an *isolated* vertex, the canonical illustration of why $V(G)$ cannot be recovered from $E(G)$.



Reading off the neighbourhood and incident-edge sets at a :

$$N_G(a) = \{b, c, e\}, \quad I_G(a) = \{\{a, b\}, \{a, c\}, \{a, e\}\}.$$

Both have size 3, consistent with $|N_G(v)| = |I_G(v)|$. The remaining vertices:

$$\begin{aligned} N_G(b) &= \{a, c\}, & I_G(b) &= \{\{a, b\}, \{b, c\}\}, \\ N_G(c) &= \{a, b\}, & I_G(c) &= \{\{a, c\}, \{b, c\}\}, \\ N_G(d) &= \emptyset, & I_G(d) &= \emptyset, \\ N_G(e) &= \{a\}, & I_G(e) &= \{\{a, e\}\}. \end{aligned}$$

5 Degree of a vertex

The number of edges touching a vertex is so fundamental that it gets its own name. Because every graph G assigns a single non-negative integer to each vertex, degree is presented as a *function* from $V(G)$ into $\mathbb{N}$ — well-defined because $V(G)$ is finite, hence so is each $I_G(v)$.

Definition 5.1 (Degree). Given a graph G , the **degree** function

$$\deg_G: V(G) \rightarrow \mathbb{N}$$

is defined, for any $v \in V(G)$, by

$$\deg_G(v) := |I_G(v)| = |N_G(v)|.$$

That is, $\deg_G(v)$ is the number of edges incident to v , equivalently the number of neighbours of v . The two counts agree because the correspondence $w \mapsto \{v, w\}$ is a bijection between $N_G(v)$ and $I_G(v)$.

Example. For the five-vertex graph from the previous section:

$$\deg_G(a) = 3, \quad \deg_G(b) = 2, \quad \deg_G(c) = 2, \quad \deg_G(d) = 0, \quad \deg_G(e) = 1.$$

The isolated vertex has degree 0. The sum of degrees is $3 + 2 + 2 + 0 + 1 = 8 = 2|E(G)|$ — each edge contributes 1 to the degree of each of its two endpoints, so it is counted twice in the total.

6 Subgraphs

A subgraph is what you get by throwing away some vertices and some edges from a graph G , while keeping the result a valid graph in its own right — that is, every remaining edge must still have both its endpoints in the remaining vertex set.

Definition 6.1 (Subgraph). A graph H is a **subgraph** of G , written $H \leq G$, if and only if

$$V(H) \subseteq V(G) \quad \wedge \quad E(H) \subseteq E(G).$$

Remark. The two inclusions are not independent. Because H must itself be a graph, every $e \in E(H)$ is a 2-element subset of $V(H)$; combined with $E(H) \subseteq E(G)$ this means we cannot keep an edge whose endpoints were dropped. So removing a vertex from $V(G)$ implicitly removes every edge incident to it.

Remark (Induced subgraphs). A particularly important construction — and the one used in the proof below — is the *induced subgraph*: choose any $S \subseteq V(G)$ and take $V(H) := S$ together with all edges of G whose endpoints both lie in S . The induced subgraph keeps as many edges as possible given its vertex set.

7 The handshaking lemma

We can now prove the identity that the previous example flagged: in every graph, the sum of the degrees equals twice the number of edges.

Theorem 7.1 (Handshaking Lemma). For any graph G ,

$$2 \cdot |E(G)| = \sum_{v \in V(G)} \deg_G(v).$$

The combinatorial intuition is one line: *every edge has two endpoints, so when we sum degrees we count each edge twice*. Turning that into a formal proof requires induction on the number of vertices, since $V(G)$ is the only object we can control by size.

Proof. We prove the equivalent statement

$$(\forall n \in \mathbb{N}) (\forall G) \left((G \text{ is a graph} \wedge |V(G)| = n) \Rightarrow 2 \cdot |E(G)| = \sum_{v \in V(G)} \deg_G(v) \right)$$

by weak induction on n — essentially, induction on the order of G .

Base case ($n = 0$). Let G be a graph with $|V(G)| = 0$, so $V(G) = \emptyset$. Then any $W \subseteq \emptyset$ satisfies $W = \emptyset$, hence $|W| = 0 \neq 2$, so

$$E(G) \subseteq \{ W \mid W \subseteq \emptyset \wedge |W| = 2 \} = \emptyset,$$

and $E(G) = \emptyset$. Therefore

$$2 \cdot |E(G)| = 2 \cdot 0 = 0 = \sum_{v \in \emptyset} \deg_G(v) = \sum_{v \in V(G)} \deg_G(v),$$

using the convention that the empty sum is 0.

Inductive step. Let $k \in \mathbb{N}$ and assume the inductive hypothesis: for every graph G with $|V(G)| = k$, $2 \cdot |E(G)| = \sum_{v \in V(G)} \deg_G(v)$. Now let G be a graph with $|V(G)| = k + 1$. Since $k + 1 > 0$, the set $V(G)$ is nonempty, so we may pick some $x \in V(G)$. We aim to remove x from G and apply the inductive hypothesis to the smaller graph that remains.

Define a graph H by

$$V(H) := V(G) \setminus \{x\}, \quad E(H) := E(G) \setminus I_G(x) = \{ e \in E(G) \mid x \notin e \}.$$

That is, H is obtained from G by deleting the vertex x and every edge incident to it. By construction H is a graph, $V(H) \subseteq V(G)$, $E(H) \subseteq E(G)$, and $|V(H)| = (k + 1) - 1 = k$, so by the inductive hypothesis

$$2 \cdot |E(H)| = \sum_{v \in V(H)} \deg_H(v). \quad (\star)$$

Step 1: degrees of vertices that are neither x nor neighbours of x are unchanged. For any $v \in V(G) \setminus (N_G(x) \cup \{x\})$, every edge e of G containing v satisfies $x \notin e$ (otherwise $\{v, x\}$ would be in $E(G)$, putting v in $N_G(x)$). Hence

$$\begin{aligned} \deg_G(v) &= |I_G(v)| \\ &= |\{ e \in E(G) \mid v \in e \}| \\ &= |\{ e \in E(G) \mid v \in e \wedge x \notin e \}| \\ &= |\{ e \in E(H) \mid v \in e \}| && \text{(by definition of } E(H)) \\ &= |I_H(v)| = \deg_H(v). \end{aligned}$$

Step 2: degrees of neighbours of x drop by exactly one. For any $v \in N_G(x)$, the edge set $I_G(v)$ splits as the disjoint union of those edges containing x and those not. There is exactly one edge in G containing both v and x , namely $\{v, x\}$, so

$$\begin{aligned} \deg_G(v) &= |I_G(v)| \\ &= |\{ e \in E(G) \mid v \in e \wedge x \in e \}| + |\{ e \in E(G) \mid v \in e \wedge x \notin e \}| \\ &= 1 + |\{ e \in E(H) \mid v \in e \}| && \text{(by definition of } E(H)) \\ &= 1 + \deg_H(v). \end{aligned}$$

Step 3: assemble the sum. The vertex set $V(G)$ partitions into three pairwise disjoint pieces:

$$V(G) = N_G(x) \cup (V(G) \setminus (N_G(x) \cup \{x\})) \cup \{x\}.$$

Splitting the sum across this partition,

$$\sum_{v \in V(G)} \deg_G(v) = \sum_{v \in N_G(x)} \deg_G(v) + \sum_{v \in V(G) \setminus (N_G(x) \cup \{x\})} \deg_G(v) + \deg_G(x).$$

Apply Step 2 to the first sum and Step 1 to the second:

$$\begin{aligned} &= \sum_{v \in N_G(x)} (1 + \deg_H(v)) + \sum_{v \in V(G) \setminus (N_G(x) \cup \{x\})} \deg_H(v) + \deg_G(x) \\ &= |N_G(x)| + \sum_{v \in N_G(x)} \deg_H(v) + \sum_{v \in V(G) \setminus (N_G(x) \cup \{x\})} \deg_H(v) + \deg_G(x). \end{aligned}$$

The two $\deg_H$ sums together range over $V(G) \setminus \{x\} = V(H)$, so they combine into $\sum_{v \in V(H)} \deg_H(v)$. Also $|N_G(x)| = |I_G(x)| = \deg_G(x)$. Therefore

$$\begin{aligned} \sum_{v \in V(G)} \deg_G(v) &= \deg_G(x) + \sum_{v \in V(H)} \deg_H(v) + \deg_G(x) \\ &= 2 \deg_G(x) + \sum_{v \in V(H)} \deg_H(v) \\ &= 2|I_G(x)| + 2|E(H)| && \text{by } (\star) \\ &= 2|\{e \in E(G) \mid x \in e\}| + 2|\{e \in E(G) \mid x \notin e\}| \\ &= 2 \cdot |E(G)|, \end{aligned}$$

since the two sets $\{e \in E(G) \mid x \in e\}$ and $\{e \in E(G) \mid x \notin e\}$ partition $E(G)$. This completes the inductive step, and the result follows by induction. $\square$

Corollary 7.2 (Parity). In any graph G , the number of vertices of odd degree is even.

Proof. By the handshaking lemma, $\sum_{v \in V(G)} \deg_G(v) = 2|E(G)|$ is even. Splitting the sum by parity of degree gives

$$\underbrace{\sum_{v: \deg_G(v) \text{ even}} \deg_G(v)}_{\text{even (sum of even numbers)}} + \sum_{v: \deg_G(v) \text{ odd}} \deg_G(v) = 2|E(G)|.$$

Subtracting the first sum (even) from $2|E(G)|$ (even) leaves the second sum even. A sum of odd numbers is even if and only if there are an even number of summands, so $|\{v : \deg_G(v) \text{ odd}\}|$ is even. $\square$

Example. For the five-vertex graph from Section 4 with degrees $(3, 2, 2, 0, 1)$, the odd-degree vertices are a (degree 3) and e (degree 1) — exactly two of them, which is even. $\checkmark$

8 Summary

1. A graph G consists of a finite vertex set $V(G)$ and an edge set $E(G) \subseteq \{W \mid W \subseteq V(G) \wedge |W| = 2\}$.
2. The cardinality constraint $|W| = 2$ forbids loops; the set structure of $E(G)$ forbids parallel edges; the unordered-set structure of each edge makes the graph undirected; finiteness of $V(G)$ forces $E(G)$ to be finite as well.
3. The *neighbourhood* $N_G(v) = \{w \in V(G) \mid \{v, w\} \in E(G)\}$ collects the vertices adjacent to v ; the *incident edge set* $I_G(v) = \{e \in E(G) \mid v \in e\}$ collects the edges touching v ; and $|N_G(v)| = |I_G(v)|$.
4. The *degree* $\deg_G: V(G) \rightarrow \mathbb{N}$ sends $v \mapsto |I_G(v)| = |N_G(v)|$.
5. $H \leq G$ (a *subgraph*) means $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$.
6. **Handshaking lemma:** $\sum_{v \in V(G)} \deg_G(v) = 2|E(G)|$, proved by induction on $|V(G)|$ via the deletion of a vertex x and its incident edges.
7. **Parity corollary:** the number of odd-degree vertices is always even.

Remark (Looking ahead). With degree, subgraphs, and handshaking established, the next questions turn *global*: when can we walk from one vertex to another following edges (*paths* and *connectivity*), what does it mean for two graphs to be “the same up to relabelling” (*isomorphism*), and which special families of graphs (trees, bipartite graphs, complete graphs) admit cleaner counts and structural theorems?